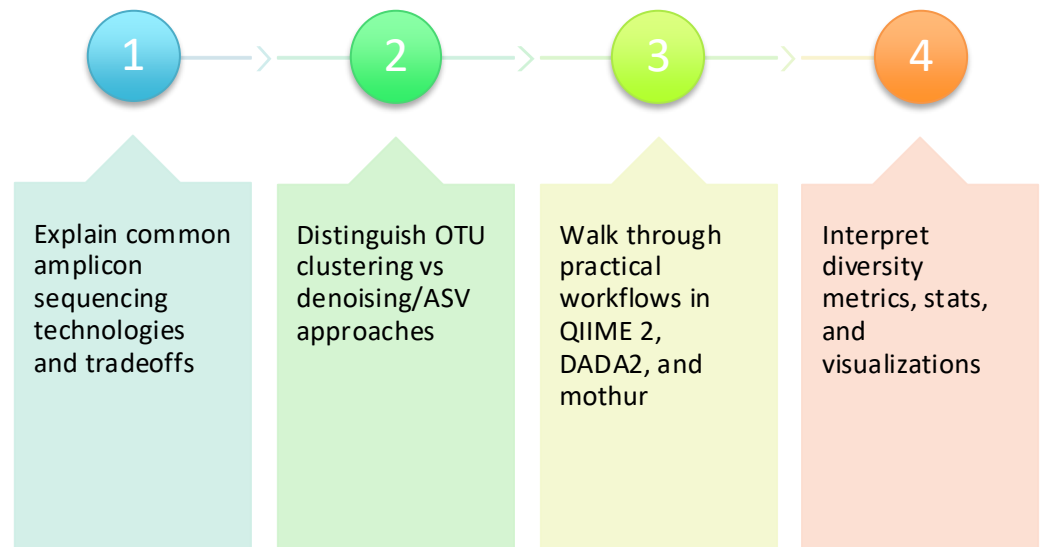


Microbial Diversity from Amplicon Data: Technologies & Analysis Workflows

BIOL6997/6993 UPRM 2026



Overview



Amplicon sequencing: overview



TARGET MARKER GENES
(16S, 18S, ITS, FUNCTIONAL
MARKERS)



PCR AMPLIFICATION →
LIBRARY PREP →
SEQUENCING



GOALS: COMMUNITY
COMPOSITION, DIVERSITY,
DIFFERENTIAL ABUNDANCE



Sequencing technologies: short-read



Illumina MiSeq /
NovaSeq: paired-end,
low raw error rate, high
throughput



Read length constraints
(e.g., 2×150–2×300)
affect overlap for
amplicons



Pros: accuracy, cost per
base; Cons: limited
read length, chimeras,
PCR bias

Sequencing
technologies:
long-read &
synthetic

PacBio HiFi: long accurate reads—full-length 16S/ITS, better phylogenetic resolution

Oxford Nanopore: very long reads, higher raw error—improving with basecallers

Synthetic long-reads (LoopSeq, 10X Genomics etc.)

Key
laboratory
confounders

Primer bias, PCR cycles,
chimera formation

Contamination (reagents,
index hopping)

Mock communities,
negative/positive controls,
randomized extraction

Analysis
paradigms:
OTUs vs
ASVs

OTU clustering: similarity threshold (97%), clusters sequences → reduced resolution

ASVs (DADA2, Deblur): exact sequence inference → higher resolution

Tradeoffs: OTUs may mask fine-scale variation; ASVs require good error modeling



Typical amplicon analysis pipeline (steps)



Demultiplexing &
primer/adaptor trimming

Quality control and filtering

Denoising or clustering (ASV or
OTU)

Chimera removal

Taxonomic assignment &
phylogenetic placement

Diversity analyses and
differential abundance



QIIME 2: overview & strengths

Plugin-based, provenance-tracked
platform; interactive visualizations (.qzv)

Supports Deblur, DADA2, OTU clustering,
taxonomy, phylogeny, diversity stats



QIIME 2: representative commands & workflow (summary)

- **Import data** -> qiime tools import
- **Trim/denoise** -> qiime dada2 denoise-paired or qiime deblur denoise-16S
- **Taxonomy** -> qiime feature-classifier classify-sklearn
- **Phylogeny & diversity** -> qiime phylogeny / qiime diversity core-metrics-phylogenetic



- Statistical error-modeling to infer exact sequences (ASVs)
- Works in R; can be run standalone or inside QIIME 2
- High resolution, built-in chimera removal

DADA2: core ideas & strengths



- `filterAndTrim()` -> `learnErrors()` -> `derepFastq()` -> `dada()` -> `makeSequenceTable()` -> `removeBimeraDenovo()`
- Assign taxonomy with `assignTaxonomy()` or external classifiers

DADA2: typical R pipeline (summary)



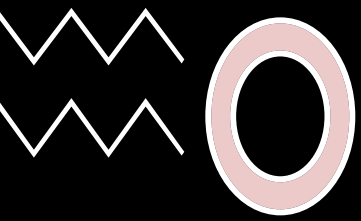
mothur: overview & strengths

- Comprehensive, scriptable pipeline; popular SOPs (Schloss lab)
- OTU and ASV-like workflows (pre.cluster), chimera detection tools

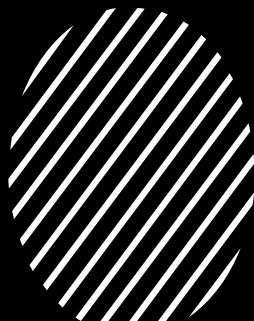


mothur: brief workflow (commands)

- `make.contigs()` ->
`screen.seqs()` -> `unique.seqs()`
-> `pre.cluster()` ->
`chimera.uchime()` ->
`classify.seqs()` -> `cluster.split()`



Taxonomic assignment & reference databases



Common DBs: SILVA, Greengenes (legacy), RDP, UNITE (fungi), GTDB



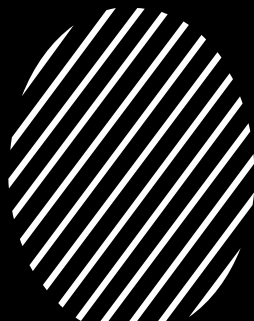
Choose database by marker region coverage, curation frequency, and taxonomic framework



Use region-specific / trimmed training sets when possible; update regularly



Diversity metrics & statistics



Alpha diversity: richness (count of taxa) — sensitive to sampling depth

Alpha diversity: Shannon (richness + evenness) — captures dominance structure

Alpha diversity: Faith's PD (phylogenetic diversity) — includes evolutionary distances

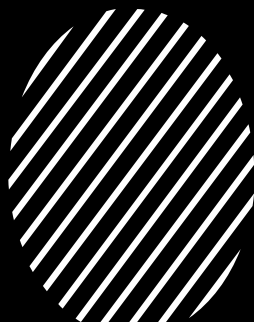
Beta diversity: Bray–Curtis (abundance-based) — sensitive to abundant taxa

Beta diversity: Jaccard (presence/absence) — focuses on shared taxa regardless of abundance

Beta diversity: UniFrac (phylogenetic; weighted/unweighted) — incorporates phylogeny and abundance



Data QC & contamination control (best practices)



Use blanks and mock communities; remove contaminants (decontam)

Rarefaction vs normalization — consider compositional approaches

Visual QC: per-sample read counts, quality plots, chimera rates

Practical tool comparison & recommendations

- QIIME 2: reproducible end-to-end pipelines
- DADA2: high-resolution ASVs and R-based flexibility
- mothur: legacy SOPs and OTU-focused workflows



Reproducibility & FAIR principles

01

Findable: deposit raw reads (SRA/ENA), provide persistent IDs and searchable metadata

02

Accessible: use standard repositories and clear access instructions (open or controlled access)

03

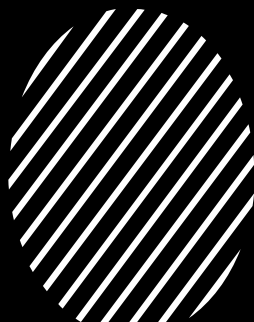
Interoperable: use standard formats (FASTQ/BIOM/TSV), controlled vocabularies, and detailed metadata

04

Reusable: provide methods, parameters, code, and license; include provenance and environment (Conda, Docker)



R packages for downstream analysis & visualization



vegan — community ecology &
ordination



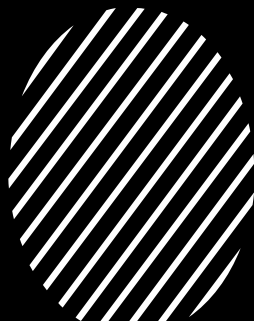
phyloseq — integrate ASV/OTU tables,
taxonomy, phylogeny, and metadata



qiime2R — import QIIME 2 artifacts
into R for phyloseq workflows



Resources & further reading



QIIME 2 docs and tutorials
(docs.qiime2.org)

DADA2 vignette
(benjjneb.github.io/dada2)

mothur SOP (mothur.org)

Practical assignment suggestion (before the workshop)

Reprocess a small MiSeq dataset (soil sample in QIIME2 tutorial): primer removal → DADA2 → taxonomy → alpha/beta → PERMANOVA



Compare results: DADA2 vs OTU clustering (mothur) on same dataset

https://github.com/sabahzero/UPRM_QIIME2workshop

March 3-4, 2026

Workshop Overview

This repository operates as a syllabus for the UPRM [QIIME 2](#) Workshop. We utilize this toolkit to process 16S & 18S reads from raw data through hypothesis testing and visualization, with aims as follows:

```
Quality control and denoising
Taxonomic assignment
Alpha and beta diversity analysis
Statistical testing and figure generation
```

This workshop will utilize data from the *C. californicus* venom microbiome study. Participants will explore:

```
Basic alpha/beta diversity analyses
Tissue differences (distal, proximal, bulb) between control and experimental samples
Population/location differences
Taxa with potential protective or symbiotic functions
Cone snail literature gaps regarding ecological considerations
```

Instructors

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Sierra Smith (In-Person) @sierra-smith
Tim Colston (In-Person) @

Key Dates

- Pre-requisites [Feb 28]: (a) Installation of required software, (b) processing of Raw Reads
- Session 1 [Mar 3]: 12-3 PM
- Session 2 [Mar 4]: 1-3 PM AST
- Additional Office Hours: 3-6 PM, both days Mar 3 & 4

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